Analysis 1

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2026/27

About these notes

These editable study notes are based on *Analysis* by Christopher Brocklebank and James Yang (October 2024), following Prof. Hadžić's UCL Analysis 1 course in 2024–25. They are a rewritten companion, rather than a verbatim transcription or official course material.

The four chapters follow the source's order. Definitions, principal results, worked examples and proof arguments have been rewritten in consistent notation. Repeated attempts at proofs have been combined.

Obvious mathematical errors have been corrected. Some arguments left unfinished or assigned as exercises in the source are supplied here; these additions are identified where relevant. The source's missing figure placeholders are not reproduced. The accompanying README records the main editorial decisions.

Throughout, $\mathbb{N} = \{1, 2, 3, \ldots\}$ and all sequences are real unless stated otherwise. A statement labelled non-examinable in the source is retained with that attribution; this is not a statement about the current year's assessment.

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Chapter 1

Foundations and Sequences

1.1 What analysis is about

Analysis provides precise definitions and proofs for ideas from calculus: limits, infinite sums, continuity and derivatives. An *axiom* is a starting assumption; a *theorem* is a statement proved from assumptions. A *lemma* is a result used to establish another result, and a *corollary* follows readily from a previous result.

Example 1.1 (Why infinite sums need definitions)**.** The harmonic series $\sum_{n=1}^{\infty} 1/n$ diverges, whereas $\sum_{n=1}^{\infty}(-1)^{n+1}/n$ converges. Rearranging a conditionally convergent series may change its sum. Thus the rules for finite sums cannot be applied to infinite sums without checking their hypotheses. Chapter 2 makes these statements precise.

Example 1.2 (Smooth does not imply equal to a Taylor series)**.** Consider

$$f(x) = \begin{cases} e^{-1/x^2}, & x > 0, \\ 0, & x \le 0. \end{cases}$$

All derivatives at 0 exist and are zero, so its Taylor series there is the zero series. Nevertheless $f(x) > 0$ for $x > 0$. Merely having derivatives of all orders does not ensure that a function equals its Taylor series. This is a motivating example; the differentiability assertion is revisited at the end of Chapter 4.

1.2 Number systems and rational approximation

1.2.1 From natural numbers to real numbers

The familiar number systems satisfy

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

Here $0 \notin \mathbb{N}$. Addition and multiplication on $\mathbb{N}$ are commutative and associative, and multiplication distributes over addition:

$$\begin{aligned} m + n &= n + m, & mn &= nm, \\ (m + n) + p &= m + (n + p), & (mn)p &= m(np), \\ m(n + p) &= mn + mp. \end{aligned}$$

For any m, n, exactly one of $m < n$, $m = n$, $m > n$ holds. Order is transitive.

Subtraction need not stay in $\mathbb{N}$: $x + m = n$ has the integer solution $x = n - m$. This motivates $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$. Division need not stay in $\mathbb{Z}$, motivating

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\}.$$

Fractions have the rules

$$\frac{m}{n} + \frac{p}{q} = \frac{mq + pn}{nq}, \qquad \frac{m}{n}\frac{p}{q} = \frac{mp}{nq}, \qquad \frac{m/n}{p/q} = \frac{mq}{np} \quad (p \neq 0).$$

Every rational has a reduced representation with coprime numerator and denominator. Terminating or eventually periodic decimal expansions represent rational numbers. Non-periodic decimal expansions motivate a larger system, $\mathbb{R}$; a real number outside $\mathbb{Q}$ is called *irrational*. We will use completeness, rather than decimals, to characterise $\mathbb{R}$.

Lemma 1.1 (Parity)**.** *If an integer n is odd, then n^2 is odd. Consequently, if n^2 is even, then n is even.*

Proof. Writing $n = 2k + 1$ gives $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. The second statement is the contrapositive. □

Theorem 1.2 (Irrationality of $\sqrt{2}$)**.** *There is no rational number x with $x^2 = 2$.*

Proof. Suppose $x = p/q$ in lowest terms, with $q > 0$. Then $p^2 = 2q^2$, so p is even. Write $p = 2k$. Substitution gives $4k^2 = 2q^2$, hence $q^2 = 2k^2$, and q is also even. This contradicts coprimality. □

1.2.2 Pell's equation and approximation

Once the positive square root of 2 is available, the equation $|p^2 - 2q^2| = 1$, for $p, q \in \mathbb{N}$, gives

$$\left| \frac{p}{q} - \sqrt{2} \right| = \frac{1}{q^2(p/q + \sqrt{2})} < \frac{1}{q^2}.$$

Thus a solution with large q gives a good rational approximation.

Proposition 1.3 (A recursive family of approximations)**.** *Define $p_1 = q_1 = 1$ and*

$$p_{n+1} = p_n + 2q_n, \qquad q_{n+1} = p_n + q_n.$$

Then $p_n, q_n \in \mathbb{N}$, $p_n^2 - 2q_n^2 = (-1)^n$, and $q_n \geq n$. Consequently $p_n/q_n \to \sqrt{2}$.

Proof. The assertions hold at $n = 1$. If they hold at n, then

$$\begin{aligned} p_{n+1}^2 - 2q_{n+1}^2 &= (p_n + 2q_n)^2 - 2(p_n + q_n)^2 \\ &= -(p_n^2 - 2q_n^2) = (-1)^{n+1}, \end{aligned}$$

and $q_{n+1} \geq 1 + n$. Induction proves the claims. The error is less than $1/q_n^2 \leq 1/n^2$, which tends to zero. □

n	p_n	q_n	p_n/q_n (approximately)
1	1	1	1.000000
2	3	2	1.500000
3	7	5	1.400000
4	17	12	1.416667
5	41	29	1.413793
6	99	70	1.414286

Remark. An induction proof checks a base case and proves that the assertion for n implies the assertion for $n+1$. The limiting conclusion above completes an argument left unfinished on source p. 9.

1.3 Ordered fields and inequalities

Definition 1.4 (Field)**.** A field F is a set with addition and multiplication satisfying the following properties for all $a, b, c \in F$:

1. $(a+b)+c = a+(b+c)$ and $a+b = b+a$.
2. There is $0 \in F$ with $a+0 = a$; every a has an additive inverse $-a$.
3. $(ab)c = a(bc)$ and $ab = ba$.
4. There is $1 \in F$, $1 \neq 0$, with $a1 = a$; every nonzero a has an inverse a^{-1} with $aa^{-1} = 1$.
5. $a(b+c) = ab + ac$.

These are the nine arithmetic axioms, grouped by operation.

Definition 1.5 (Ordered field)**.** An ordered field is a field with a total order $\leq$ satisfying antisymmetry and transitivity, together with

$$a \leq b \implies a + c \leq b + c, \qquad a \leq b,\ c \geq 0 \implies ac \leq bc.$$

Totality means that $a \leq b$ or $b \leq a$ for every pair. Multiplication by a negative number reverses the inequality.

Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. The condition $c \geq 0$ is essential; it was omitted from the corresponding axiom in the source.

Lemma 1.6 (Basic field consequences)**.** *In a field, $a0 = 0$. Also, $ab = 0$ implies $a = 0$ or $b = 0$.*

Proof. Distributivity gives $a0 = a(0+0) = a0 + a0$; subtract $a0$. If $a \neq 0$, multiply $ab = 0$ by a^{-1} to obtain $b = 0$. □

Lemma 1.7. *For $0 \leq a, b$ in an ordered field, $a \leq b$ if and only if $a^2 \leq b^2$.*

Proof. If $a \leq b$, then $a^2 \leq ab \leq b^2$. Conversely, $a > b \geq 0$ would give $a^2 - b^2 = (a-b)(a+b) > 0$, contradicting $a^2 \leq b^2$. □

Definition 1.8 (Absolute value)**.** For $x \in \mathbb{R}$,

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

The quantity $|x - y|$ is the distance between x and y.

Useful identities include $|xy| = |x||y|$, $|x|^2 = x^2$, and $|x| < r \iff -r < x < r$ for $r > 0$.

Proposition 1.9 (Triangle and reverse triangle inequalities)**.** *For $x, y \in \mathbb{R}$,*

$$|x + y| \leq |x| + |y|, \qquad \big||x| - |y|\big| \leq |x - y|.$$

Proof. Since $xy \leq |x||y|$, we have $(x+y)^2 \leq (|x|+|y|)^2$. Both sides to be compared after taking square roots are nonnegative, giving the triangle inequality. Applying it to $x = (x-y)+y$ gives $|x| - |y| \leq |x-y|$. Interchanging x, y gives the other half of the reverse inequality. □

1.4 Completeness, supremum and infimum

Definition 1.10 (Bounds and extremal elements)**.** A set $S \subseteq \mathbb{R}$ is *bounded above* if there is $M \in \mathbb{R}$ with $x \leq M$ for every $x \in S$. It is *bounded below* if there is m with $m \leq x$ for every $x \in S$. It is *bounded* if both hold.

A maximum is an upper bound belonging to S; a minimum is a lower bound belonging to S. Write $\max S$ and $\min S$ when they exist.

Definition 1.11 (Supremum and infimum)**.** For a nonempty set bounded above, $s = \sup S$ means:

1. $x \leq s$ for every $x \in S$;
2. for every $\varepsilon > 0$, some $x \in S$ satisfies $s - \varepsilon < x$.

Thus s is the least upper bound. For a nonempty set bounded below, $t = \inf S$ means that $t \leq x$ for all $x \in S$ and that every $\varepsilon > 0$ admits some $x \in S$ with $x < t + \varepsilon$.

Theorem 1.12 (Completeness of the real numbers)**.** *There exists a complete ordered field, denoted $\mathbb{R}$. Completeness means that every nonempty subset bounded above has a supremum in $\mathbb{R}$. Equivalently, every nonempty subset bounded below has an infimum in $\mathbb{R}$.*

We take existence of this field as a foundational result; its construction is outside these notes. The infimum statement follows from $\inf S = -\sup(-S)$. Completeness is the extra property distinguishing $\mathbb{R}$ from $\mathbb{Q}$.

Proposition 1.13. *If $\max S$ exists, then $\sup S = \max S$. Similarly, if $\min S$ exists, then $\inf S = \min S$.*

Proof. A maximum $s \in S$ is an upper bound, and every upper bound is at least s because it bounds every element of S. The minimum proof is analogous. □

Theorem 1.14 (Archimedean property)**.** *The natural numbers are not bounded above in $\mathbb{R}$.*

Proof. If $H = \sup \mathbb{N}$ existed, $H - 1$ would not be an upper bound. Hence some $n \in \mathbb{N}$ would satisfy $n > H - 1$, and $n + 1 > H$ would contradict that H bounds $\mathbb{N}$. □

In particular, for every $\varepsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \varepsilon$.

Example 1.3 (Existence of $\sqrt{2}$ from completeness)**.** Let $S = \{x \in \mathbb{R} : x > 0,\ x^2 < 2\}$. It contains 1 and is bounded above by 2, so $M = \sup S$ exists, with $1 \leq M \leq 2$. We show $M^2 = 2$.

If $M^2 < 2$, choose $0 < h < 1$ so that $(2M + 1)h < 2 - M^2$. Then

$$(M + h)^2 = M^2 + 2Mh + h^2 < M^2 + (2M + 1)h < 2,$$

contradicting that M is an upper bound.

If $M^2 > 2$, choose $0 < h < M$ with $2Mh < M^2 - 2$. Then $(M - h)^2 > M^2 - 2Mh > 2$. Every $x \in S$ must satisfy $x < M - h$, since otherwise $x^2 \geq (M - h)^2 > 2$. Thus $M - h$ is an upper bound smaller than M, another contradiction. Therefore $M^2 = 2$, and M is the positive square root.

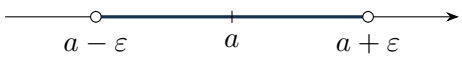
Example 1.4 (Intervals and endpoints)**.** For $a < b$, the intervals are

$$(a, b) = \{x : a < x < b\}, \quad [a, b] = \{x : a \leq x \leq b\}, \quad [a, b) = \{x : a \leq x < b\}.$$

Set	Infimum	Minimum	Supremum	Maximum
(a, b)	a	none	b	none
$[a, b]$	a	a	b	b
$[a, b)$	a	a	b	none
$[a, \infty)$	a	a	none in $\mathbb{R}$	none

Example 1.5. For $S = \{3 - 1/n : n \in \mathbb{N}\}$, $\inf S = \min S = 2$ and $\sup S = 3$, while no maximum exists. Indeed, all its elements are less than 3. Given $\varepsilon > 0$, choose $n > 1/\varepsilon$; then $3 - 1/n > 3 - \varepsilon$. The same argument gives $\sup\{1 - 1/n : n \in \mathbb{N}\} = 1$.

Definition 1.15 (Neighbourhood)**.** For $a \in \mathbb{R}$ and $\varepsilon > 0$, the ε-neighbourhood of a is $(a - \varepsilon, a + \varepsilon) = \{x : |x - a| < \varepsilon\}$.



1.5 Sequences and their limits

Definition 1.16 (Sequence and boundedness)**.** A real sequence is a function $\mathbb{N} \to \mathbb{R}$, written $(x_n)_{n \geq 1}$. Its range is $\{x_n : n \in \mathbb{N}\}$. A sequence is bounded above, below, or on both sides when its range has the corresponding property. Equivalently, it is bounded if some $M > 0$ satisfies $|x_n| \leq M$ for all n.

For example, $x_n = n$ is bounded below but not above, whereas $x_n = (-1)^n$ is bounded. A finite initial part of a sequence does not determine its limit.

Definition 1.17 (Convergence)**.** We say $x_n \to L \in \mathbb{R}$ if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N : \quad |x_n - L| < \varepsilon.$$

We also write $\lim_{n \to \infty} x_n = L$. A sequence that has no finite real limit is called divergent.

The order of quantifiers matters: N may depend on ε, but must work for *every* subsequent n. Replacing $n \geq N$ by $n > N$, or allowing a real threshold N, gives an equivalent definition.

Example 1.6 (Direct limit proofs)**.** For $x_n = 1/n$, given $\varepsilon > 0$, choose $N > 1/\varepsilon$. Then $n \geq N$ implies $|x_n| \leq 1/N < \varepsilon$, so $1/n \to 0$.

Also $1 - 2^{-n} \to 1$, since $2^n \geq n$ and $|(1 - 2^{-n}) - 1| = 2^{-n} \leq 1/n$. For another example,

$$\left| \frac{2n^2 - 1}{n^2 + 1} - 2 \right| = \frac{3}{n^2 + 1} \leq \frac{3}{n^2}.$$

Choosing $N > \sqrt{3/\varepsilon}$ proves the limit is 2.

Remark (Negating convergence to a specified value). The statement $x_n \not\to L$ means

$$\exists \varepsilon_0 > 0 \; \forall N \in \mathbb{N} \; \exists n \geq N : \quad |x_n - L| \geq \varepsilon_0.$$

For example, $1/n \not\to 1$: choose $\varepsilon_0 = 1/2$, and take $n \geq \max\{N, 2\}$. To prove divergence, one must exclude *every* possible finite limit, not merely one proposed value.

Theorem 1.18 (Uniqueness of the limit)**.** *A sequence has at most one limit.*

Proof. Suppose $x_n \to L_1$ and $x_n \to L_2$, with $L_1 \neq L_2$. Set $\varepsilon = |L_1 - L_2|/2$. For sufficiently large n, both distances to x_n are less than ε. Thus

$$|L_1 - L_2| \leq |L_1 - x_n| + |x_n - L_2| < 2\varepsilon = |L_1 - L_2|,$$

a contradiction. □

Theorem 1.19 (Convergent sequences are bounded)**.** *If $x_n \to L \in \mathbb{R}$, then (x_n) is bounded.*

Proof. Use $\varepsilon = 1$ to find N with $|x_n - L| < 1$ for $n \geq N$. Then $|x_n| < |L| + 1$ on this tail. The finitely many earlier terms are bounded as well: take $M = \max\{|x_1|, \ldots, |x_N|, |L| + 1\}$. □

Theorem 1.20 (Sandwich theorem)**.** *Suppose $x_n \leq y_n \leq z_n$ for all sufficiently large n and both $x_n \to L$ and $z_n \to L$. Then $y_n \to L$.*

Proof. For any $\varepsilon > 0$, sufficiently large n satisfy $L - \varepsilon < x_n \leq y_n \leq z_n < L + \varepsilon$. □

Example 1.7. Since $-1/n^2 \leq (-1)^n/n^2 \leq 1/n^2$, the sequence $(-1)^n/n^2$ tends to 0.

1.6 Algebra of sequence limits

Theorem 1.21 (Algebra of limits)**.** *If $x_n \to x$ and $y_n \to y$, then*

$$x_n + y_n \to x + y, \qquad cx_n \to cx, \qquad x_n y_n \to xy.$$

If additionally $y \neq 0$, then $y_n \neq 0$ eventually and $x_n/y_n \to x/y$ on that tail. If all $y_n \neq 0$, the quotient is a sequence from the outset.

Proof. **Sum.** Make $|x_n - x| < \varepsilon/2$ and $|y_n - y| < \varepsilon/2$. The triangle inequality gives $|x_n + y_n - (x + y)| < \varepsilon$. The constant-multiple rule follows directly from $|cx_n - cx| = |c||x_n - x|$ (and is immediate for $c = 0$).

Product. By Theorem 1.19, choose $M \geq 1$ with $|y_n| \leq M$ for all n. For sufficiently large n, both errors $|x_n - x|$ and $|y_n - y|$ are less than $\varepsilon/(M + |x|)$. Then

$$\begin{aligned} |x_n y_n - xy| &= |y_n(x_n - x) + x(y_n - y)| \\ &\leq M|x_n - x| + |x||y_n - y| < \varepsilon. \end{aligned}$$

Quotient. Since $y \neq 0$, eventually $|y_n - y| < |y|/2$, so $|y_n| > |y|/2$. Hence

$$\left|\frac{1}{y_n} - \frac{1}{y}\right| = \frac{|y_n - y|}{|y_n||y|} \leq \frac{2|y_n - y|}{|y|^2} \longrightarrow 0.$$

Apply the product rule to $x_n(1/y_n)$. □

Remark. If $y_n \neq 0$ for every n and $y_n \to y \neq 0$, then in fact $|y_n| \geq c > 0$ for every n. Take the minimum of $|y|/2$ and the finitely many nonzero initial absolute values. This is the source's auxiliary quotient-rule lemma.

Example 1.8. Dividing numerator and denominator by n^2 gives

$$\frac{2n^2 - 1}{n^2 + 1} = \frac{2 - 1/n^2}{1 + 1/n^2} \longrightarrow 2.$$

This shorter argument becomes valid once the algebra of limits is proved.

Proposition 1.22 (Order passes to limits)**.** *If $x_n \leq y_n$ eventually, $x_n \to x$, and $y_n \to y$, then $x \leq y$.*

Proof. If $x > y$, choose $\varepsilon = (x - y)/3$. Eventually $x_n > x - \varepsilon > y + \varepsilon > y_n$, a contradiction. □

1.7 Infinite limits and monotone sequences

Definition 1.23 (Divergence to infinity)**.** We write $x_n \to +\infty$ if for every $M \in \mathbb{R}$ some N satisfies $x_n > M$ whenever $n \geq N$. We write $x_n \to -\infty$ if for every $m \in \mathbb{R}$ eventually $x_n < m$.

These are forms of divergence, not convergence to a real number. For example, $n^2 \to +\infty$, but $(-1)^n$ has neither a finite nor an infinite limit.

Definition 1.24 (Monotonicity)**.** A sequence is *increasing* (nondecreasing) if $x_{n+1} \geq x_n$ for all n, and *strictly increasing* if these inequalities are strict. Decreasing and strictly decreasing are defined by reversing the signs. A sequence is *monotone* if it is increasing or decreasing.

Theorem 1.25 (Monotone convergence)**.** *An increasing sequence bounded above converges to* $\sup\{x_n : n \in \mathbb{N}\}$*. A decreasing sequence bounded below converges to its infimum.*

Proof. In the increasing case, put $s = \sup\{x_n\}$. Given $\varepsilon > 0$, some $x_N > s - \varepsilon$. For every $n \geq N$, $s - \varepsilon < x_N \leq x_n \leq s < s + \varepsilon$. Apply this result to $(-x_n)$ to obtain the decreasing case. □

An increasing sequence unbounded above tends to $+\infty$: once it exceeds any prescribed M, every subsequent term does too. The decreasing analogue holds for $-\infty$.

Example 1.9 (A recursive limit)**.** Let $x_1 = 1$ and $x_{n+1} = (x_n + 1)/3$. Induction gives $x_n \geq 1/2$, because $x_n \geq 1/2$ implies $x_{n+1} \geq (1/2 + 1)/3 = 1/2$. Also

$$x_{n+1} - x_n = \frac{2}{3}\left(\frac{1}{2} - x_n\right) \leq 0.$$

The sequence is decreasing and bounded below, so it converges to some L. Passing to the limit gives $L = (L + 1)/3$, hence $L = 1/2$.

Remark (Prove existence before solving for the limit)*.* For the Fibonacci sequence $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$, blindly assuming a finite limit gives $L = 2L$, hence $L = 0$. This is invalid: the sequence is unbounded. Indeed $F_{n+2} \geq F_{n+1} + 1$, so it grows without bound. A fixed-point equation identifies a possible limit only after convergence has been justified.

1.8 Subsequences and Bolzano–Weierstrass

Definition 1.26 (Subsequence)**.** If $j_1 < j_2 < \cdots$ are natural numbers, then (x_{j_n}) is a subsequence of (x_n). Necessarily $j_n \geq n$.

Theorem 1.27. *If $x_n \to L$, every subsequence tends to L. The same holds for $L = +\infty$ or $-\infty$.*

Proof. Once all terms with index at least N satisfy the required estimate, x_{j_n} satisfies it for $n \geq N$, since $j_n \geq n$. □

Thus two subsequences with different limits prove divergence. In particular, a fixed shift x_{n+k} has the same limit as x_n.

Example 1.10. For $x_n = 1/n + \sin(n\pi/2)$, we have $x_{2n} = 1/(2n) \to 0$ while $x_{4n+1} = 1/(4n + 1) + 1 \to 1$. Therefore (x_n) diverges.

Lemma 1.28. *Every real sequence has a monotone subsequence.*

Proof. Call n a *peak* if $x_n \geq x_m$ for every $m > n$. If there are infinitely many peaks, list them in increasing order $j_1 < j_2 < \cdots$. Then (x_{j_n}) is decreasing.

If there are only finitely many peaks, choose j_1 beyond all of them (or $j_1 = 1$ if there are none). As j_1 is not a peak, some $j_2 > j_1$ satisfies $x_{j_2} > x_{j_1}$. Repeating constructs a strictly increasing subsequence. □

Theorem 1.29 (Bolzano–Weierstrass)**.** *Every bounded real sequence has a convergent subsequence.*

Proof. Take a monotone subsequence using the lemma. It is still bounded, so it converges by Theorem 1.25. □

Example 1.11 (Many subsequential limits)**.** List the fractions in $(0,1)$ in successive denominator blocks:

$$\frac{1}{2}; \quad \frac{1}{3}, \frac{2}{3}; \quad \frac{1}{4}, \frac{2}{4}, \frac{3}{4}; \quad \dots.$$

Every $r \in [0,1]$ is the limit of a subsequence. For each denominator $q \geq 2$, choose $p_q \in \{1, \dots, q-1\}$ with $|p_q/q - r| \leq 1/q$. Picking this entry from each consecutive block gives a subsequence tending to r. This supplies the exercise argument on source p. 32.

1.9 The Cauchy–Schwarz inequality

Theorem 1.30 (Cauchy–Schwarz)**.** *For real $x_1, \dots, x_n$ and $y_1, \dots, y_n$,*

$$\left(\sum_{k=1}^{n} x_k y_k \right)^2 \leq \left(\sum_{k=1}^{n} x_k^2 \right) \left(\sum_{k=1}^{n} y_k^2 \right).$$

Proof. Write $A = \sum y_k^2$, $B = \sum x_k y_k$ and $C = \sum x_k^2$. If $A = 0$, every $y_k = 0$ and the claim is immediate. Otherwise,

$$0 \leq \sum_{k=1}^{n} (x_k - t y_k)^2 = A t^2 - 2Bt + C$$

for every real t. Substituting $t = B/A$ gives $C - B^2/A \geq 0$, proving $B^2 \leq AC$. □

For $n = 2$, the difference between the right and left sides is $(x_1 y_2 - x_2 y_1)^2$. In $\mathbb{R}^3$, the geometric interpretation is $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$, since the dot product equals $\|\mathbf{x}\| \|\mathbf{y}\| \cos\theta$.

Chapter 2

Infinite Series

2.1 Partial sums and convergence

Definition 2.1 (Infinite series)**.** For a sequence (a_n), define its partial sums by

$$S_N = \sum_{n=1}^{N} a_n.$$

The series $\sum_{n=1}^{\infty} a_n$ converges to $S \in \mathbb{R}$ if $S_N \to S$. Otherwise the series diverges. A series is therefore defined through a limit of finite sums.

Example 2.1 (A warning about grouping)**.** The series $1 - 1 + 1 - 1 + \cdots$ has partial sums $1, 0, 1, 0, \ldots$, so it diverges. Grouping it as $(1 - 1) + (1 - 1) + \cdots$ or as $1 + (-1 + 1) + \cdots$ does not assign a sum to the original divergent series.

Proposition 2.2 (Basic properties)**.** *Suppose $\sum a_n = S$ and $\sum b_n = T$. Then:*

1. *$a_n \to 0$.*
2. *The tails satisfy $\sum_{n=k}^{\infty} a_n \to 0$ as $k \to \infty$.*
3. *For $c \in \mathbb{R}$, $\sum ca_n = cS$.*
4. *$\sum(a_n + b_n) = S + T$.*

Proof. Since $a_n = S_n - S_{n-1}$ for $n \geq 2$, the first claim follows from the algebra of limits. The tail from k is $S - S_{k-1}$, which tends to zero. The last two claims follow by applying the constant-multiple and sum rules to the corresponding finite partial sums. The tail argument and sum-rule details complete exercises left to the reader in the source. □

Remark. The condition $a_n \to 0$ is necessary, but not sufficient, for convergence. The harmonic series below is the standard counterexample. Changing, adding or deleting finitely many terms does not affect convergence, although it can change the value of the sum.

2.2 Geometric and telescoping series

Example 2.2 (Geometric series)**.** For $r \neq 1$,

$$\sum_{n=0}^{N} r^n = \frac{1 - r^{N+1}}{1 - r}.$$

If $|r| < 1$, then $r^N \to 0$, hence

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}, \qquad \sum_{n=1}^{\infty} r^n = \frac{r}{1-r}.$$

For completeness, if $0 < |r| < 1$, put $|r|^{-1} = 1+h$, $h > 0$. Bernoulli's inequality $(1+h)^n \geq 1+nh$ gives $|r|^n \leq 1/(1+nh) \to 0$. Bernoulli's inequality follows by induction: multiplying $1+nh$ by $1+h$ gives at least $1+(n+1)h$. If $|r| \geq 1$, the terms r^n do not tend to zero, so the geometric series diverges.

Example 2.3 (Telescoping)**.** Using $1/[n(n+1)] = 1/n - 1/(n+1)$,

$$\begin{aligned}\sum_{n=1}^{N} \frac{1}{n(n+1)} &= \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{N+1} \longrightarrow 1.\end{aligned}$$

Thus $\sum_{n=1}^{\infty} 1/[n(n+1)] = 1$.

2.3 Comparison and absolute convergence

For nonnegative terms $a_n \geq 0$, the partial sums are increasing. They either are bounded and converge, or are unbounded and tend to $+\infty$. This observation is the basis of comparison tests.

Theorem 2.3 (Comparison test)**.** *If $0 \leq a_n \leq b_n$ for all sufficiently large n and $\sum b_n$ converges, then $\sum a_n$ converges. If instead $\sum a_n$ diverges, then $\sum b_n$ diverges. When the inequalities hold for every n, $\sum a_n \leq \sum b_n$ in the convergent case.*

Proof. First suppose the inequalities hold for every n. The partial sums $A_N = \sum_{n=1}^{N} a_n$ are increasing and satisfy $A_N \leq B_N \leq \sum b_n$. Thus they converge by monotone convergence. The sum inequality follows by passing to limits. If comparison holds only after a finite index, apply this argument to the tails, since finitely many initial terms do not affect convergence. The divergence claim is the contrapositive. □

This includes the source's "improved comparison test": eventual comparison is sufficient.

Example 2.4. Since $0 \leq 2^n/(3^n+1) \leq (2/3)^n$, the series $\sum 2^n/(3^n+1)$ converges. Also, for $n \geq 2$,

$$0 < \frac{1}{n^3-2} \leq \frac{2}{n^3}.$$

Once convergence of $\sum 1/n^3$ has been established, comparison proves convergence of $\sum_{n=1}^{\infty} 1/(n^3-2)$. Its first term is -1; comparison is applied only from $n = 2$ onward.

Definition 2.4 (Absolute and conditional convergence)**.** A series $\sum a_n$ is *absolutely convergent* if $\sum |a_n|$ converges. It is *conditionally convergent* if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Theorem 2.5. *Every absolutely convergent real series converges.*

Proof. Put $a_n^+ = \max\{a_n, 0\}$ and $a_n^- = \max\{-a_n, 0\}$. Then $a_n = a_n^+ - a_n^-$, $|a_n| = a_n^+ + a_n^-$, and $0 \leq a_n^{\pm} \leq |a_n|$. Both $\sum a_n^+$ and $\sum a_n^-$ converge by comparison. Subtracting their partial-sum limits proves convergence of $\sum a_n$. □

Theorem 2.6 (p-series)**.** *For $p > 0$, the series $\sum_{n=1}^{\infty} n^{-p}$ converges if and only if $p > 1$.*

Proof. For $0 < p \le 1$, $n^{-p} \ge 1/n$. The harmonic partial sums obey

$$\sum_{n=1}^{2^M} \frac{1}{n} = 1 + \sum_{j=1}^{M} \sum_{n=2^{j-1}+1}^{2^j} \frac{1}{n} \ge 1 + \frac{M}{2}.$$

Each block contains 2^{j-1} terms, each at least 2^{-j}. Thus the harmonic series, and consequently each series with $0 < p \le 1$, diverges to $+\infty$.

For $p > 1$, group terms with $2^j \le n < 2^{j+1}$:

$$\sum_{n=2^j}^{2^{j+1}-1} \frac{1}{n^p} \le \frac{2^j}{(2^j)^p} = (2^{1-p})^j.$$

The geometric series with ratio $2^{1-p} < 1$ gives a common bound for all partial sums. Since the partial sums increase, they converge. □

For irrational p, real powers are defined later using the exponential and logarithm. The same estimates apply to those powers.

Theorem 2.7 (Limit comparison)**.** *Let $a_n, b_n > 0$ eventually and suppose $a_n/b_n \to \ell$, where $0 < \ell < \infty$. Then $\sum a_n$ converges if and only if $\sum b_n$ converges.*

Proof. Eventually $\ell/2 < a_n/b_n < 3\ell/2$, so compare both ways with positive constant multiples of b_n. This proof supplies the tutorial result cited, without proof, in the source. □

2.4 Alternating series

Theorem 2.8 (Alternating series test)**.** *If $a_n \ge 0$, $a_{n+1} \le a_n$, and $a_n \to 0$, then $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges.*

Proof. Let $S_N = \sum_{n=1}^{N} (-1)^{n+1} a_n$. The even partial sums increase:

$$S_{2N+2} - S_{2N} = a_{2N+1} - a_{2N+2} \ge 0.$$

The odd partial sums decrease: $S_{2N+1} - S_{2N-1} = -a_{2N} + a_{2N+1} \le 0$. Also $S_{2N} \le S_{2N-1} \le a_1$ and $S_{2N} \ge 0$. Thus both subsequences converge, and their limits agree because $S_{2N-1} - S_{2N} = a_{2N} \to 0$. Every sufficiently late partial sum is an even or odd term close to that common limit, so the entire sequence of partial sums converges. □

Example 2.5. The alternating harmonic series $1 - 1/2 + 1/3 - 1/4 + \cdots$ converges by this test, but not absolutely because $\sum 1/n$ diverges. Its sum is $\log 2$; that identity is mentioned in the source but not established by the alternating series test alone.

2.5 Ratio and root tests

Theorem 2.9 (Ratio test)**.** *Suppose $a_n \neq 0$ eventually and $|a_{n+1}|/|a_n| \to \ell$. Then:*

1. *If $\ell < 1$, $\sum a_n$ converges absolutely.*
2. *If $\ell > 1$, $|a_n| \to +\infty$, so $\sum a_n$ diverges.*
3. *If $\ell = 1$, the test gives no conclusion.*

The divergence conclusion also holds when the ratio tends to $+\infty$.

Proof. For $\ell < 1$, choose r with $\ell < r < 1$. Eventually $|a_{n+1}| \leq r|a_n|$, and induction gives $|a_{N+k}| \leq |a_N|r^k$. Comparison with a geometric series proves absolute convergence. For $\ell > 1$, choose $1 < r < \ell$. Eventually the reverse bound gives $|a_{N+k}| \geq |a_N|r^k \to +\infty$. □

Example 2.6 (The exponential series)**.** Define, for every real x,

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad 0! = 1.$$

For $x \neq 0$, the ratio of successive absolute terms is $|x|/(n+1) \to 0$, so the series converges absolutely. For $x = 0$, the sum is just 1. This defines exp before its continuity or differentiability is known.

Theorem 2.10 (Root test)**.** *Suppose $\sqrt[n]{|a_n|} \to \ell$. If $\ell < 1$, $\sum a_n$ converges absolutely. If $\ell > 1$, it diverges. If $\ell = 1$, the test is inconclusive.*

Proof. For $\ell < 1$, choose $\ell < r < 1$. Eventually $|a_n| \leq r^n$, so comparison proves absolute convergence. For $\ell > 1$, choose $1 < r < \ell$. Eventually $|a_n| \geq r^n$, so the terms do not tend to zero. □

Example 2.7 (Recognising inconclusive cases)**.** For both $a_n = 1/n$ and $a_n = 1/n^2$, the successive ratios tend to 1. The first series diverges and the second converges, so ratio limit 1 cannot decide convergence.

Example 2.8 (Using the root test)**.** For $a_n = (-1)^n((2n+7)/(5n+2))^n$,

$$\sqrt[n]{|a_n|} = \frac{2n+7}{5n+2} \longrightarrow \frac{2}{5} < 1.$$

Hence $\sum a_n$ converges absolutely.

2.5.1 Choosing a convergence test

First check whether $a_n \to 0$. If not, the series diverges. For nonnegative terms, try comparison or limit comparison with a geometric or p-series. Factorials often suggest the ratio test; an expression raised to the nth power often suggests the root test. For terms with changing signs, test absolute convergence, or use the alternating series test when its monotonicity and zero-limit hypotheses hold.

2.6 Rearrangements

Definition 2.11. A rearrangement of $\sum_{n=1}^{\infty} a_n$ is $\sum_{n=1}^{\infty} a_{\pi(n)}$, where $\pi : \mathbb{N} \to \mathbb{N}$ is a bijection. Every original term appears exactly once.

Theorem 2.12 (Absolute convergence and rearrangement)**.** *Every rearrangement of an absolutely convergent series converges to the same sum.*

Supplementary proof. Let $S = \sum a_n$ and choose K with $\sum_{n>K} |a_n| < \varepsilon$. For sufficiently large N, the indices $\pi(1), \ldots, \pi(N)$ include $1, \ldots, K$. After removing the terms common to the rearranged partial sum and S, only some terms with indices greater than K remain. Thus

$$\left| S - \sum_{j=1}^{N} a_{\pi(j)} \right| \leq \sum_{n>K} |a_n| < \varepsilon.$$

The source states this theorem without its proof. □

Theorem 2.13 (Riemann rearrangement theorem)**.** *If a real series converges conditionally, then for every $L \in \mathbb{R}$ it has a rearrangement converging to L. There are also rearrangements whose partial sums tend to $+\infty$ or $-\infty$.*

Supplementary proof outline. The positive terms have sum $+\infty$ and the negative terms have sum $-\infty$; otherwise convergence of their difference would force absolute convergence. To obtain a finite target L, take unused positive terms in their original order until the partial sum exceeds L, then unused negative terms until it falls below L, and repeat. Each crossing error is bounded by the last term added, which tends to zero. Both lists are exhausted, and any zero terms can be inserted, so this is a rearrangement converging to L.

For $+\infty$, at stage k insert the next unused negative term, then enough unused positive terms to make the sum exceed k. At least one positive term is used at each stage. The negative terms tend to zero, so the intervening drops also tend to zero, and the partial sums tend to $+\infty$. Reverse the roles to obtain $-\infty$. □

Example 2.9 (Rearranging the alternating harmonic series)**.** Let $S = 1 - 1/2 + 1/3 - 1/4 + \cdots$. Taking two positive terms for each negative term gives

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \cdots .$$

Its sum is $3S/2$, not S. To justify this, at the end of its Nth block its sum is

$$T_{3N} = \sum_{k=1}^{2N} \frac{1}{2k-1} - \sum_{k=1}^{N} \frac{1}{2k} = S_{4N} + \frac{1}{2} \sum_{k=N+1}^{2N} \frac{1}{k}.$$

The last sum equals S_{2N}, since both equal $H_{2N} - H_N$ with $H_n = \sum_{k=1}^{n} 1/k$. Hence $T_{3N} \to S + S/2$. The one or two extra terms between block endpoints tend to zero, proving the claimed limit for all partial sums.

2.7 Cauchy products and the exponential addition law

Definition 2.14 (Cauchy product)**.** For series indexed from 0, define

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

The series $\sum_{n=0}^{\infty} c_n$ is their Cauchy product. Its terms collect products along the diagonals $k + (n-k) = n$.

Theorem 2.15. *If $\sum_{n=0}^{\infty} a_n = A$ and $\sum_{n=0}^{\infty} b_n = B$ both converge absolutely, their Cauchy product converges absolutely and has sum AB.*

Supplementary proof. Let $A_* = \sum |a_n|$ and $B_* = \sum |b_n|$. For each N,

$$\sum_{n=0}^{N} |c_n| \le \sum_{k+j \le N} |a_k||b_j| \le A_* B_*.$$

Thus the product series converges absolutely. Compare its Nth partial sum with the rectangular product

$$\left(\sum_{k=0}^{N} a_k \right) \left(\sum_{j=0}^{N} b_j \right).$$

Their difference consists of terms with $0 \le k, j \le N$ and $k + j > N$. Each such term has either $k > N/2$ or $j > N/2$. Its total absolute value is therefore at most

$$B_* \sum_{k>N/2} |a_k| + A_* \sum_{j>N/2} |b_j| \longrightarrow 0.$$

The rectangular product tends to AB, proving the result. The source omits this proof. □

Corollary 2.16 (Exponential addition law)**.** *For all $x, y \in \mathbb{R}$,* $\exp(x+y) = \exp(x)\exp(y)$.

Proof. Both exponential series converge absolutely. Their nth Cauchy-product coefficient is, by the finite binomial theorem,

$$\sum_{k=0}^{n} \frac{x^k y^{n-k}}{k!(n-k)!} = \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} = \frac{(x+y)^n}{n!}.$$

Summing gives the identity. □

In particular, $\exp(0) = 1$ and $\exp(x)\exp(-x) = 1$. Moreover $\exp(x) > 0$: it is positive for $x \ge 0$ by its series, and for $x < 0$ it is the reciprocal of the positive number $\exp(-x)$.

Chapter 3

Limits of Functions and Continuity

3.1 Limits and one-sided limits

A function limit describes values near a point. Its value at that point can be different, or the function may not be defined there at all.

Definition 3.1 (Limit at an interior point)**.** Suppose f is defined on an open interval about c, except possibly at c. We write $\lim_{x\to c} f(x) = L$ if

$$\forall \varepsilon > 0\ \exists \delta > 0 : \quad 0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon.$$

The condition $0 < |x - c|$ excludes the point c itself.

Definition 3.2 (One-sided limits)**.** The right-hand limit $\lim_{x\to a^+} f(x) = L$ means that for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$a < x < a + \delta \implies |f(x) - L| < \varepsilon,$$

for x in the domain. The left-hand limit at b uses $b - \delta < x < b$. We also write $x \downarrow a$ and $x \uparrow b$, respectively.

At an interior point a two-sided limit exists with value L exactly when both one-sided limits exist and equal L.

Example 3.1 (Different one-sided limits)**.** Let

$$f(x) = \begin{cases} 2x, & x < 1, \\ 1, & x = 1, \\ 4 - x, & x > 1. \end{cases}$$

Then $\lim_{x\to 1^-} f(x) = 2$ and $\lim_{x\to 1^+} f(x) = 3$, so the two-sided limit does not exist. For the left limit, $|f(x) - 2| = 2|x - 1|$, so choose $\delta = \varepsilon/2$. For the right limit, $|f(x) - 3| = |x - 1|$, so choose $\delta = \varepsilon$. The assigned value $f(1) = 1$ does not affect either limit.

Example 3.2 (Oscillation controlled by a small factor)**.** For $x \neq 0$, let $f(x) = 1 + x^2 \cos(1/x)$. Since

$$|f(x) - 1| \leq x^2,$$

choosing $\delta = \sqrt{\varepsilon}$ proves $\lim_{x\to 0} f(x) = 1$. No value of $f(0)$ was needed.

3.2 Continuity

Definition 3.3 (Continuity on a domain)**.** Let $S \subseteq \mathbb{R}$ and $c \in S$. A function $f : S \to \mathbb{R}$ is continuous at c if

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in S : \quad |x - c| < \delta \implies |f(x) - f(c)| < \varepsilon.$$

It is continuous on S, written $f \in C(S)$, if it is continuous at every point of S. Otherwise it is discontinuous at any point where the condition fails.

At an interior point of an interval, continuity is equivalent to $\lim_{x \to c} f(x) = f(c)$. On $[a, b]$, the endpoint conditions are

$$\lim_{x \to a^+} f(x) = f(a), \qquad \lim_{x \to b^-} f(x) = f(b).$$

Only points in the domain are tested.

Example 3.3 (A removable discontinuity)**.** Extend $1 + x^2 \cos(1/x)$ to 0 by setting $f(0) = \gamma$. The extension is continuous at 0 if and only if $\gamma = 1$.

Example 3.4 (Thomae's function)**.** On $(0, 1)$ define

$$T(x) = \begin{cases} 0, & x \notin \mathbb{Q}, \\ 1/q, & x = p/q \text{ in lowest terms, } q > 0. \end{cases}$$

Then $\lim_{x \to c} T(x) = 0$ for every $c \in (0, 1)$. It is continuous at every irrational point and discontinuous at every rational point.

Supplementary explanation. Given $\varepsilon > 0$, choose an integer $Q > 1/\varepsilon$. There are only finitely many reduced fractions in $(0, 1)$ with denominator at most Q. A sufficiently small punctured neighbourhood of c avoids all of them, other than possibly c itself. At any other rational point in that neighbourhood, $q > Q$, so $T(x) < \varepsilon$; at an irrational point $T(x) = 0$. Therefore the punctured limit is zero. This equals $T(c)$ exactly when c is irrational.

3.3 Sequential criteria and algebra

Theorem 3.4 (Sequential criterion for limits)**.** *Under the domain assumptions for an interior limit, $\lim_{x \to c} f(x) = L$ if and only if every sequence (x_n) in the domain with $x_n \neq c$ and $x_n \to c$ satisfies $f(x_n) \to L$. For a one-sided limit, require the sequence to approach from that side.*

Proof. Suppose the limit exists. Given $\varepsilon > 0$, choose a corresponding $\delta > 0$. Since $x_n \to c$, eventually $0 < |x_n - c| < \delta$, and therefore $|f(x_n) - L| < \varepsilon$.

Conversely, if the limit condition fails, there is $\varepsilon_0 > 0$ such that for every $\delta > 0$ there is a domain point x satisfying $0 < |x - c| < \delta$ and $|f(x) - L| \geq \varepsilon_0$. Choose such a point x_n for $\delta = 1/n$. Then $x_n \to c$, but $f(x_n) \not\to L$, contradicting the sequential hypothesis. Restrict the selected points to the appropriate side for one-sided limits. □

Corollary 3.5 (Sequential continuity)**.** *A function $f : S \to \mathbb{R}$ is continuous at $c \in S$ if and only if for every sequence $x_n \in S$ tending to c, we have $f(x_n) \to f(c)$.*

The proof is the same, without requiring $x_n \neq c$.

Theorem 3.6 (Algebra of function limits)**.** *If $f(x) \to L$ and $g(x) \to M$ as $x \to c$, then*

$$f(x) + g(x) \to L + M, \qquad f(x)g(x) \to LM.$$

If $M \neq 0$, then g is nonzero sufficiently near c and $f(x)/g(x) \to L/M$. The analogous rules hold for one-sided limits. In particular, sums and products of functions continuous at c are continuous there; their quotient is continuous if $g(c) \neq 0$.

Proof. Apply the sequence limit rules to $f(x_n)$ and $g(x_n)$ for every sequence approaching c, then use Theorem 3.4. Nonvanishing follows by making $|g(x) - M| < |M|/2$. □

Theorem 3.7 (Sandwich theorem for functions)**.** *If $f(x) \leq g(x) \leq h(x)$ sufficiently near c, and both bounding functions tend to L as $x \to c$, then $g(x) \to L$. Continuity of g is not assumed.*

Proof. Apply the sequence sandwich theorem to $f(x_n), g(x_n), h(x_n)$ for any sequence approaching c, or use the same $L \pm \varepsilon$ bounds directly. □

Example 3.5. For $x > 0$, $-\sqrt{x} \leq \sqrt{x}\cos(1/x) \leq \sqrt{x}$. Both bounds tend to zero as $x \to 0^+$, so $\sqrt{x}\cos(1/x) \to 0$.

Example 3.6 (Standard continuous functions)**.** Constant functions and $f(x) = x$ are continuous directly from the definition. The algebra rules show that every polynomial is continuous, and every rational function $p(x)/q(x)$ is continuous wherever $q(x) \neq 0$. For instance $(x^2 - 2x + 7)/(x^2 + 1)$ is continuous on $\mathbb{R}$.

Theorem 3.8 (Composition)**.** *If g is continuous at c and f is continuous at $g(c)$, with the composition defined near c in its domain, then $f \circ g$ is continuous at c.*

Proof. For $x_n \to c$, continuity gives $g(x_n) \to g(c)$, then $f(g(x_n)) \to f(g(c))$. □

3.4 Continuity of the exponential

Recall the definition $\exp(x) = \sum_{n=0}^{\infty} x^n/n!$ and the addition law from Chapter 2. We already know positivity and $\exp(-x) = 1/\exp(x)$.

Lemma 3.9 (Bounds near zero)**.** *For $0 \leq x < 1$,*

$$1 + x \leq \exp(x) \leq \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

For $-1 < x < 0$, writing $t = -x \in (0, 1)$ and taking reciprocals gives

$$1 + x \leq \exp(x) \leq \frac{1}{1-x}.$$

Hence the same bounds hold throughout $|x| < 1$.

Proof. For nonnegative x, use positivity of the exponential terms and $1/n! \leq 1$. For $x = -t$, reciprocate $1 + t \leq \exp(t) \leq 1/(1-t)$, using positivity. □

This argument supplies local bounds without assuming any continuity in advance. The stronger inequality $\exp(x) \geq 1 + x$ for all real x is cited as homework in the source; these local bounds suffice here.

Theorem 3.10. *The exponential function is continuous on* $\mathbb{R}$.

Proof. As $x \to 0$, both $1 + x$ and $1/(1 - x)$ tend to 1. The sandwich theorem gives $\exp(x) \to 1 = \exp(0)$. For any fixed c,

$$\exp(c + h) = \exp(c)\exp(h) \longrightarrow \exp(c) \quad (h \to 0),$$

which proves continuity at c. □

For example, $\exp((x^3 - 7x)/(3x^2 + 5))$ is continuous on all of $\mathbb{R}$ by composition.

3.5 Continuous functions on a closed bounded interval

Lemma 3.11. *If* $x_n \in [a, b]$ *and* $x_n \to x$, *then* $x \in [a, b]$.

Proof. Pass to limits in $a \le x_n \le b$ using preservation of order under limits. □

Together with Bolzano–Weierstrass, this shows every sequence in $[a, b]$ has a subsequence converging to a point still in $[a, b]$. This property is called *sequential compactness*.

Theorem 3.12 (Boundedness)**.** *Every continuous function* $f : [a, b] \to \mathbb{R}$ *is bounded.*

Proof. If it were unbounded, choose $x_n \in [a, b]$ with $|f(x_n)| > n$. A subsequence x_{j_n} converges to $c \in [a, b]$ by compactness. Continuity implies $f(x_{j_n}) \to f(c)$, hence this image subsequence is bounded. But $|f(x_{j_n})| > j_n \ge n$, a contradiction. □

Theorem 3.13 (Extreme value theorem)**.** *Every continuous function* $f : [a, b] \to \mathbb{R}$ *attains both a maximum and a minimum: there exist* $x_{\max}, x_{\min} \in [a, b]$ *with*

$$f(x_{\max}) = \sup f([a, b]), \qquad f(x_{\min}) = \inf f([a, b]).$$

Proof. Boundedness gives $M = \sup f([a, b]) \in \mathbb{R}$. For each n choose $x_n \in [a, b]$ with $M - 1/n < f(x_n) \le M$. A subsequence tends to $c \in [a, b]$; its image tends to $f(c)$ by continuity and to M by the sandwich theorem. Therefore $f(c) = M$. Apply the same argument to $-f$ to obtain the minimum. This completes the proof left unfinished on source p. 59. □

Example 3.7 (Why the hypotheses matter)**.** The continuous function $1/x$ on $(0, 1]$ is unbounded: the interval is not closed. The continuous function x on $[0, \infty)$ is unbounded: this interval is closed but not bounded. Finally, on the compact interval $[0, 1]$, setting $f(0) = 1$ and $f(x) = 1/x$ for $x > 0$ gives an unbounded function because continuity at 0 fails.

3.6 The intermediate value theorem

Lemma 3.14 (Preservation of a strict inequality)**.** *If* f *is continuous at* c *and* $f(c) > \lambda$, *then* $f(x) > \lambda$ *for all domain points sufficiently near* c. *An analogous statement holds for* $f(c) < \lambda$.

Proof. Choose $\varepsilon = (f(c) - \lambda)/2 > 0$ and require $|f(x) - f(c)| < \varepsilon$. Then $f(x) > f(c) - \varepsilon > \lambda$. Apply this to $-f$ for the other case. □

Theorem 3.15 (Intermediate value theorem)**.** *Let* $f : [a, b] \to \mathbb{R}$ *be continuous. Every real number* λ *between* $f(a)$ *and* $f(b)$ *equals* $f(c)$ *for some* $c \in [a, b]$.

Proof. The endpoint cases are immediate. Replacing f by $-f$ if necessary, assume $f(a) < \lambda < f(b)$. Set $S = \{x \in [a,b] : f(x) < \lambda\}$ and $c = \sup S$. Continuity at the endpoints gives points of S strictly larger than a and excludes a whole left neighbourhood of b from S. Consequently $a < c < b$.

If $f(c) < \lambda$, the lemma gives a point $x > c$ close to c with $f(x) < \lambda$, contradicting that c bounds S. If $f(c) > \lambda$, a neighbourhood $(c-\delta, c+\delta)$ has $f(x) > \lambda$. Since S contains no point above c either, $c - \delta$ would be an upper bound for S, after shrinking δ if necessary. This contradicts minimality of c. Therefore $f(c) = \lambda$. □

Combining the extreme and intermediate value theorems gives

$$f([a,b]) = [m, M], \qquad m = \min f([a,b]), \quad M = \max f([a,b]).$$

In particular a continuous function taking opposite signs at the endpoints has a zero between them.

3.7 The logarithm and real powers

Definition 3.16 (Bijection and inverse)**.** A function $f : A \to B$ is injective if different inputs give different outputs, and surjective if every $y \in B$ has an input $x \in A$. If both hold it is bijective and has an inverse $f^{-1} : B \to A$ with $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

Theorem 3.17. *The exponential is a strictly increasing bijection* $\exp : \mathbb{R} \to (0, \infty)$.

Proof. For $h > 0$, its series gives $\exp(h) \geq 1 + h > 1$. Thus if $y = x + h$,

$$\exp(y) = \exp(x)\exp(h) > \exp(x).$$

It is therefore injective. Also $\exp(x) \geq 1 + x \to +\infty$ as $x \to +\infty$, and $\exp(x) = 1/\exp(-x) \to 0$ as $x \to -\infty$. Given $u > 0$, choose $x_1 < x_2$ with $\exp(x_1) < u < \exp(x_2)$. Continuity and the intermediate value theorem give x with $\exp(x) = u$, proving surjectivity. □

Definition 3.18 (Natural logarithm)**.** The inverse of exp is the natural logarithm $\log = \ln : (0, \infty) \to \mathbb{R}$. In particular,

$$\exp(\log x) = x, \qquad \log(\exp t) = t.$$

Define $e = \exp(1) = \sum_{n=0}^{\infty} 1/n! \approx 2.718281828459$.

Definition 3.19 (Real powers)**.** For $a > 0$ and $x \in \mathbb{R}$, define $a^x = \exp(x \log a)$.

This agrees with the usual definition for rational exponents: for positive integers q, $(a^{p/q})^q = a^p$, and positivity selects the positive root. The addition law gives, for $a > 0$ and $x, y \in \mathbb{R}$,

$$a^{x+y} = a^x a^y, \qquad (a^x)^y = a^{xy}.$$

Also $e^x = \exp(x)$. If $a \neq 1$, the map $x \mapsto a^x$ is a bijection $\mathbb{R} \to (0, \infty)$, with inverse $\log_a$. For $a = 1$ it is constant. For a negative base, a real-valued a^x cannot be defined for every real x with these same conventions.

Chapter 4

Differentiation

4.1 Derivatives and linear approximation

For the line $y = ax + b$, the slope is $(y_2 - y_1)/(x_2 - x_1) = a$. For a general graph, consider the secant through $(c, f(c))$ and $(c + h, f(c + h))$. Its slope is $[f(c + h) - f(c)]/h$. The derivative is the limiting slope as the second point approaches the first.

Definition 4.1 (Derivative)**.** Let f be defined on an open interval containing c. It is differentiable at c if the finite limit

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists. Differentiability on an open interval means differentiability at every point there. Another notation is $\left.\frac{df}{dx}\right|_{x=c}$.

Example 4.1 (Derivatives from the definition)**.** For $f(x) = p$ constant, the difference quotient is zero, so $f' = 0$. For $f(x) = x$, it equals 1, so $f' = 1$. As an additional example, for $f(x) = x^2$ it equals

$$\frac{(c+h)^2 - c^2}{h} = 2c + h \longrightarrow 2c.$$

The quadratic calculation replaces the source's incomplete third example.

Theorem 4.2 (Linear approximation)**.** *The function f is differentiable at c if and only if some $\mu \in \mathbb{R}$ and remainder $R(h)$ satisfy, for sufficiently small h,*

$$f(c + h) = f(c) + \mu h + hR(h), \qquad R(h) \longrightarrow 0 \quad (h \to 0).$$

In that case $\mu = f'(c)$.

Proof. If $f'(c)$ exists, put $\mu = f'(c)$ and $R(h) = [f(c+h) - f(c)]/h - \mu$ for $h \neq 0$, with $R(0) = 0$. Then the identity holds and $R(h) \to 0$. Conversely, subtract $f(c)$ and divide by $h \neq 0$ in the asserted identity. The quotient is $\mu + R(h) \to \mu$, proving differentiability. □

The affine expression $f(c) + f'(c)h$ is the linear approximation. Its error $hR(h)$ is smaller than h in the precise sense that the ratio tends to zero. The condition $R(h) \to 0$ is essential.

Theorem 4.3 (Differentiability implies continuity)**.** *If f is differentiable at c, it is continuous at c.*

Proof. Linear approximation gives $f(c + h) - f(c) = h(f'(c) + R(h)) \to 0$. □

The converse fails: $|x|$ is continuous at 0, but its left and right difference quotients are -1 and 1.

4.2 Rules for differentiation

Theorem 4.4 (Sum, constant multiple and product rules)**.** *If f, g are differentiable at c and $a \in \mathbb{R}$, then*

$$\begin{aligned}(f+g)'(c) &= f'(c) + g'(c),\\ (af)'(c) &= af'(c),\\ (fg)'(c) &= f'(c)g(c) + f(c)g'(c).\end{aligned}$$

Proof. For sums and constant multiples, expand the difference quotients and use the algebra of limits. For products, add and subtract $f(c)g(c+h)$:

$$\frac{f(c+h)g(c+h) - f(c)g(c)}{h} = \frac{f(c+h)-f(c)}{h}\, g(c+h) + f(c)\frac{g(c+h)-g(c)}{h}.$$

Continuity gives $g(c+h) \to g(c)$, so passing to limits gives the product rule. This shortens the source's equivalent remainder expansion. □

Theorem 4.5 (Sandwich theorem for derivatives)**.** *Suppose f, q are differentiable at c, $f(c) = q(c) = L$, and $f'(c) = q'(c) = m$. If $f(x) \le g(x) \le q(x)$ near c, including at c, then g is differentiable at c and $g'(c) = m$.*

Proof. First $g(c) = L$. For $h > 0$, subtract L and divide by h to sandwich the difference quotient for g between those for f and q. Both tend to m. For $h < 0$, the inequalities reverse, but both outer limits are still m. Thus both one-sided limits of the middle quotient equal m. □

Theorem 4.6 (Chain rule)**.** *If g is differentiable at c and f is differentiable at $g(c)$, with the composition defined near c, then*

$$(f \circ g)'(c) = f'(g(c))g'(c).$$

Proof. By linear approximation,

$$g(c+h) - g(c) = h(g'(c) + R_g(h)) =: k(h), \qquad R_g(h) \to 0.$$

Hence $k(h) \to 0$. Also

$$f(g(c)+k) = f(g(c)) + f'(g(c))k + kR_f(k), \qquad R_f(k) \to 0.$$

Set $R_f(0) = 0$ so substitution remains valid even when $k(h) = 0$. For $h \neq 0$, the composite difference quotient becomes

$$\frac{f(g(c+h)) - f(g(c))}{h} = (g'(c) + R_g(h))\big(f'(g(c)) + R_f(k(h))\big).$$

Passing to the limit proves the formula without dividing by $k(h)$. □

4.3 Powers, exponentials and inverse functions

Example 4.2 (Integer powers)**.** Induction and the product rule give $(x^n)' = nx^{n-1}$ for $n \in \mathbb{N}$. The base case is $(x)' = 1$. If it holds for n, then

$$(x^{n+1})' = (xx^n)' = x^n + x(nx^{n-1}) = (n+1)x^n.$$

For $x \neq 0$, the reciprocal has difference quotient

$$\frac{1/(x+h) - 1/x}{h} = -\frac{1}{x(x+h)} \longrightarrow -\frac{1}{x^2}.$$

The chain rule applied to $1/x^{-n}$ gives the same power rule for negative integer n, on $x \neq 0$. The derivative of $x^0 = 1$ is zero.

Proposition 4.7 (Derivative of the exponential)**.** *For every $x \in \mathbb{R}$, $(\exp x)' = \exp x$.*

Proof. For $|h| < 1$, Lemma 3.9 gives $1 + h \leq \exp(h) \leq 1/(1-h)$. Both bounding functions equal 1 at 0 and have derivative 1 there. The derivative sandwich theorem gives $\exp'(0) = 1$. At arbitrary x, the addition law yields

$$\frac{\exp(x+h) - \exp(x)}{h} = \exp(x)\frac{\exp(h) - 1}{h} \longrightarrow \exp(x).$$

□

Corollary 4.8. *For $a > 0$, $\frac{d}{dx}a^x = a^x \log a$.*

Proof. Use $a^x = \exp(x \log a)$ and the chain rule. □

Proposition 4.9 (Derivative of an inverse: conditional formula)**.** *If f is bijective and differentiable at c, and f^{-1} is differentiable at $f(c)$, then $f'(c) \neq 0$ and*

$$(f^{-1})'(f(c)) = \frac{1}{f'(c)}.$$

Proof. Differentiate $f^{-1}(f(x)) = x$ at c. The chain rule gives $(f^{-1})'(f(c))f'(c) = 1$. □

This calculation assumes differentiability of the inverse. The inverse function theorem at the end of the chapter supplies sufficient conditions.

Example 4.3 (Logarithms and real powers)**.** Applying the inverse function theorem to exp, whose derivative is positive, justifies differentiability of log. Consequently, for $x > 0$,

$$\frac{d}{dx}\log x = \frac{1}{\exp(\log x)} = \frac{1}{x}.$$

For $\alpha \in \mathbb{R}$, $x^\alpha = \exp(\alpha \log x)$ on $x > 0$, so

$$\frac{d}{dx}x^\alpha = x^\alpha \frac{\alpha}{x} = \alpha x^{\alpha-1}.$$

For arbitrary real α, the restriction $x > 0$ matters. For instance, $\sqrt{x}$ has no finite right derivative at 0 because $\sqrt{h}/h = 1/\sqrt{h} \to +\infty$ as $h \to 0^+$.

4.4 Extrema and stationary points

Definition 4.10 (Global and local extrema)**.** A global maximum of $f : S \to \mathbb{R}$ at $c \in S$ means that $f(x) \leq f(c)$ for all $x \in S$. A global minimum reverses the inequality. A local maximum means that the inequality holds for all domain points with $|x - c| < \delta$, for some $\delta > 0$. A local minimum is analogous. An extremum is a maximum or a minimum.

For two-sided derivative tests, c must be an interior point of the domain. A stationary point is one where $f'(c) = 0$, called a critical point in the source. Some texts use "critical point" more broadly to include points where the derivative is undefined.

Theorem 4.11 (Fermat's stationary-point theorem)**.** *If f is differentiable at an interior point c where it has a local extremum, then $f'(c) = 0$.*

Proof. At a local maximum, $f(c + h) - f(c) \leq 0$ for small h. For $h > 0$, division and the limit give $f'(c) \leq 0$. For $h < 0$, they give $f'(c) \geq 0$. Hence $f'(c) = 0$. Apply the result to $-f$ for a minimum. □

The converse fails: x^3 has derivative zero at 0 but no extremum there. Endpoint extrema need not have derivative zero: $f(x) = x$ on $[-1, 1]$ has its minimum at -1 and maximum at 1. To find extrema on a closed interval, compare interior candidates with endpoint values.

4.5 Rolle's theorem and the mean value theorem

Theorem 4.12 (Rolle's theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, with $a < b$, then some $c \in (a, b)$ has $f'(c) = 0$.*

Proof. By the extreme value theorem, f attains a maximum and minimum. If the values agree, f is constant. Otherwise at least one extreme value differs from the common endpoint value, so is attained at an interior point. Its derivative is zero by Fermat's theorem. □

Theorem 4.13 (Mean value theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b), some $c \in (a, b)$ satisfies*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Put $m = [f(b) - f(a)]/(b - a)$ and $g(x) = f(x) - mx$. Then $g(a) = g(b)$, so Rolle's theorem gives $g'(c) = 0$ for some $c \in (a, b)$. Since $g' = f' - m$, the result follows. □

Corollary 4.14 (Derivative sign and monotonicity)**.** *Under the mean value theorem hypotheses, $f' > 0$ throughout (a, b) implies f is strictly increasing on $[a, b]$, and $f' < 0$ implies it is strictly decreasing. If $f' = 0$ throughout (a, b), f is constant. Non-strict derivative inequalities give non-strict monotonicity.*

Proof. For $a \leq x < y \leq b$, apply the theorem on $[x, y]$ to get $f(y) - f(x) = f'(c)(y - x)$. The positive factor $y - x$ preserves the relevant sign. □

4.6 First, second and higher derivative tests

Theorem 4.15 (First derivative test)**.** *Let f be continuous near c and differentiable on each side. If for some $\delta > 0$,*

$$f'(x) > 0 \ (c - \delta < x < c), \qquad f'(x) < 0 \ (c < x < c + \delta),$$

then c is a strict local maximum. Reversing the two signs gives a strict local minimum.

Proof. The mean value theorem on $[x, c]$ for $x < c$ gives $f(c) > f(x)$; on $[c, x]$ for $x > c$ it gives $f(x) < f(c)$. Thus c is a strict maximum. Apply this to $-f$ for the minimum case. □

Theorem 4.16 (Second derivative test)**.** *Let f be twice continuously differentiable near c, with $f'(c) = 0$. If $f''(c) > 0$, then c is a strict local minimum. If $f''(c) < 0$, then c is a strict local maximum. If $f''(c) = 0$, the test is inconclusive.*

Proof. If $f''(c) > 0$, continuity keeps f'' positive near c. Thus f' is strictly increasing there and, since $f'(c) = 0$, is negative on the left and positive on the right. Use the first derivative test. Apply this to $-f$ when $f''(c) < 0$. □

Theorem 4.17 (Higher derivative test)**.** *Suppose f is n times continuously differentiable near c, $n \geq 2$, with $f'(c) = \cdots = f^{(n-1)}(c) = 0$ and $f^{(n)}(c) \neq 0$. For odd n, c is not a local extremum. For even n, c is a strict local minimum if $f^{(n)}(c) > 0$ and a strict local maximum if $f^{(n)}(c) < 0$.*

Supplementary proof. Assume $f^{(n)}(c) > 0$. It stays positive near c. Repeated applications of the mean value theorem, using the vanishing lower derivatives at c, show that to the right of c all lower positive-order derivatives are positive and $f(x) - f(c) > 0$. To the left, the signs alternate: $f^{(n-1)}(x) < 0$, $f^{(n-2)}(x) > 0$, and ultimately $f(x) - f(c)$ has sign $(-1)^n$. For even n, both sides are above $f(c)$; for odd n they are on opposite sides. The negative case follows by applying this reasoning to $-f$. □

The source marks this result non-examinable for its 2024–25 course and omits the proof.

Example 4.4 (Vanishing derivatives)**.** For $f(x) = x^3$, $f'(0) = f''(0) = 0$ and $f'''(0) = 6$: there is no extremum at 0. By contrast, x^4 has a strict minimum there, and $-x^4$ has a strict maximum. Thus $f''(c) = 0$ does not decide the issue.

The flat function

$$F(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0 \end{cases}$$

has $F^{(k)}(0) = 0$ for every $k \geq 0$, yet $F(x) > 0$ for $x \neq 0$. It has a strict minimum although no first nonzero derivative exists.

Supplementary explanation. For every integer $m \geq 0$, $|x|^{-m} e^{-1/x^2} \to 0$: put $t = 1/x^2$ and use $e^t \geq t^j/j!$ for an integer $j > m/2$. Repeated differentiation away from zero produces $P_k(1/x)e^{-1/x^2}$ for some polynomial P_k. These expressions and their difference quotients at zero tend to zero by that estimate. Induction proves smoothness and all stated derivatives. The one-sided flat function from Chapter 1 has the same property at zero.

4.7 The inverse function theorem

This theorem gives a sufficient condition for a differentiable local inverse. A zero derivative does not preclude invertibility: x^3 is bijective on $\mathbb{R}$, although its derivative at zero is zero. Its inverse $x^{1/3}$ has no finite derivative there.

Theorem 4.18 (One-dimensional inverse function theorem)**.** *Let f be continuously differentiable on an open interval about x_0, with $f'(x_0) \neq 0$. Some open interval J about x_0 is mapped bijectively onto an open interval I about $f(x_0)$. The inverse is differentiable, and for all $y \in I$,*

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

In particular, $(f^{-1})'(f(x_0)) = 1/f'(x_0)$.

Supplementary proof. Suppose $f'(x_0) > 0$; otherwise apply the argument to $-f$. Continuity gives a neighbourhood where $f' > 0$. Choose $u < x_0 < v$ so that $[u, v]$ lies in that neighbourhood. The mean value theorem makes f strictly increasing there. By the intermediate value theorem it maps $J = (u, v)$ bijectively onto $I = (f(u), f(v))$.

Let $g = f^{-1}$. To prove its continuity at $y \in I$, set $x = g(y)$. For every sufficiently small $\varepsilon > 0$, strict increase gives $f(x - \varepsilon) < y < f(x + \varepsilon)$. All $z \in I$ sufficiently close to y lie between those two values, so $x - \varepsilon < g(z) < x + \varepsilon$. Hence $g(z) \to g(y)$. For $z \neq y$, put $x' = g(z) \neq x$. Then $x' \to x$ and

$$\frac{g(z) - g(y)}{z - y} = \left(\frac{f(x') - f(x)}{x' - x} \right)^{-1} \longrightarrow \frac{1}{f'(x)}.$$

This proves the claimed differentiability and formula. □

The source marks this material non-examinable for its 2024–25 course and leaves its proof incomplete. The added proof avoids assuming the inverse differentiable before that fact is established.